A. Decentralised Urban Traffic Management

Main objective: Design a peer-to-peer traffic control and routing system that reduces congestion, improves incident response and preserves priority access for emergency services without a central dispatcher.

Example: A mid-sized city experiences recurring congestion, unpredictable incidents and occasional event surges that overwhelm fixed signal plans. Using SPADE, simulate a network of intersection controllers, roadside sensors, connected vehicles and specialist units (tow trucks, incident responders) that exchange compact messages to detect local congestion, verify incidents, adapt signal timing and suggest route changes. The scenario could model realistic delays, sensor noise, and the impact of rerouting on neighbouring corridors so students can explore trade-offs between local optimisation and network-level effects.

B. Decentralised Smart Microgrid Management

Main objective: Create a decentralised microgrid where prosumers, storage and flexible loads coordinate peer-to-peer to balance supply and demand, trade energy locally and maintain stability during grid disturbances.

Example: A residential neighbourhood with rooftop PV, household batteries, electric vehicles and flexible appliances must operate economically and resiliently. Using SPADE, implement prosumer agents, battery controllers, local market facilitators and a grid-interface agent to negotiate short-term energy trades, schedule demand response and coordinate islanding when the main grid is lost. The scenario could include variable generation, price signals, and stability constraints (voltage/frequency proxies) so students can test market designs against operational safety.

C. Decentralised Wildlife Monitoring and Anti-Poaching Patrols

Main objective: Build a decentralised monitoring and response system that detects poaching threats, verifies events through corroboration and coordinates patrols and community responses without a central authority.

Example: A protected reserve is instrumented with acoustic sensors, camera traps, ranger radios and community informants. Using SPADE, simulate sensor agents that report events (gunshots, vehicle noise, human presence), mobile patrol agents (drones, vehicles), ranger team agents and informant agents. Include false positives, intermittent connectivity and limited patrol resources so students must design verification, reputation and task allocation mechanisms that balance speed and accuracy.

D. Decentralised Humanitarian Supply Chain and Logistics

Main objective: Design a decentralised logistics network that coordinates warehouses, transport units, scouts and distribution points to deliver aid efficiently under disrupted infrastructure.

Example: Following a natural disaster, roads are partially blocked and communications intermittent. Using SPADE, model warehouse agents, transport agents (trucks, drones), scout agents that probe route status, and distribution point agents that post demand. The simulation could include capacity limits, loading/unloading times, fuel constraints and priority classes (medical, food, shelter) so students can evaluate negotiation and rerouting strategies under stress.

E. Decentralised Urban Waste Collection and Recycling

Main objective: Implement a decentralised system that optimises waste collection, reduces overflows and improves recycling rates through local negotiation and adaptive routing.

Example: Smart bins report fill levels and contamination likelihood; collection trucks, transfer stations and recycling facilities coordinate pickups and routing using SPADE. The scenario could model stochastic waste generation (including event surges), contamination detection errors and vehicle constraints so students can design bidding, routing and contamination verification protocols that trade off cost, timeliness and recycling yield.

F. Decentralised Public Health Outbreak Detection and Response

Main objective: Create a decentralised surveillance and response system that detects outbreaks early, corroborates signals peer-to-peer and coordinates local testing and vaccination while protecting privacy.

Example: Clinics, laboratories, mobile response teams, pharmacies and citizen sensors exchange aggregated symptom counts, test requests and capacity updates using SPADE. The scenario could include lab delays, noisy citizen reports and limited test/vaccine stocks so students must design anomaly detection, corroboration windows and resource allocation protocols that balance lead time, false positives and privacy.

G. Decentralised Urban Air Quality Monitoring and Response

Main objective: Develop a decentralised network that detects pollution hotspots, verifies sources and coordinates local mitigation actions such as targeted monitoring or temporary controls.

Example: Fixed sensors, mobile monitors, municipal response units and citizen reporters exchange pollutant summaries, wind estimates and confidence scores via SPADE. The scenario could model dispersion effects, sensor drift and temporal variability so students can explore spatial-temporal fusion, reputation weighting and decentralised tasking of mobile monitors or municipal actions.

H. Decentralised Coastal Fisheries Monitoring and Sustainable Harvesting

Main objective: Design a decentralised fisheries system that monitors stocks, negotiates short-term quotas and detects illegal fishing while supporting fair local markets.

Example: Vessel agents, sensor buoys, cooperative agents and market nodes exchange catch reports, effort logs and environmental indicators using SPADE. The scenario could include migratory stock dynamics, market demand fluctuations and enforcement constraints so students can design quota negotiation, reputation systems and lightweight audit trails that encourage compliance and sustainability.

I. Decentralised Last Mile Microtransit and Ride Sharing

Main objective: Create a decentralised microtransit system that matches riders and drivers, supports pooling and repositions vehicles to improve service quality and equity.

Example: Rider agents, driver agents and microhub agents negotiate trip matches, pooling arrangements and vehicle repositioning using SPADE. The scenario could model time windows, detour tolerances, driver shifts and traffic variability so students can evaluate matching algorithms, pooling heuristics and pricing schemes that balance wait time, occupancy and fairness.

J. Decentralised Urban Parking and Curbside Management

Main objective: Build a decentralised curb management system that allocates short-term access, reduces cruising and enforces rules locally while accommodating priority uses.

Example: Curb node agents, vehicle and delivery agents, microhubs and enforcement units negotiate reservations, resolve conflicts and adapt to events using SPADE. The scenario could include arrival uncertainty, enforcement delays and competing priorities (deliveries, pickups, transit stops) so students can design reservation protocols, preemption rules and discovery mechanisms that are robust and fair.

K. Decentralised Food Rescue and Redistribution

Main objective: Design a decentralised food rescue network that matches surplus to need, manages perishability and maintains food safety without a central dispatcher.

Example: Donor agents (restaurants, supermarkets), rescuer agents (volunteer drivers), storage nodes (community fridges) and recipient agents coordinate offers, pickups and deliveries using SPADE. The scenario could model perishability, cold chain constraints, last-minute cancellations and volunteer availability so students can design perishability-aware routing, verification and reputation systems that maximise usable rescues.

L. Decentralised Drone Inspection and Rapid Repair Coordination

Main objective: Implement a decentralised inspection and repair coordination system where drones, sensors and crews detect faults, verify anomalies and perform lightweight repairs or escalate to human teams.

Example: Heterogeneous drone agents, fixed sensor nodes and repair-team agents inspect bridges, streetlights and power lines using SPADE. The scenario could include weather effects, no-fly zones, battery and tool constraints and probabilistic repair success so students can design verification chains, capability matching and safe handoff protocols that respect operational limits.

M. Decentralised Urban Flood Early Warning and Response

Main objective: Create a decentralised flood early warning and response system that detects rising water, verifies risk and coordinates local evacuation and mitigation actions.

Example: Water sensor agents, neighbourhood coordinators, volunteer responder agents, shelter agents and citizen agents collaborate in SPADE to detect rapid water-level rises, corroborate evidence and allocate pumps, sandbags and evacuation assistance. The scenario could model sensor noise, blocked drains and route closures so students can design verification windows, alert propagation and shelter matching that balance false alarms and timely action.

N. Decentralised Urban Water Network Monitoring and Leak Response

Main objective: Design a decentralised water network system that detects leaks, isolates faults and coordinates repairs while minimising service disruption and water loss.

Example: Flow and pressure sensor agents, valve controllers, maintenance crew agents and consumer agents exchange compact summaries and proposals using SPADE. The scenario could include hydraulic delays, sensor drift and network topology so students can design corroboration logic, decentralised isolation negotiation and demand-side coordination to stabilise pressure and reduce losses.

O. Decentralised Community Emergency Medical Response

Main objective: Build a decentralised emergency medical response system that detects time-critical incidents, verifies them locally and dispatches nearest capable responders while guiding bystanders and preserving privacy.

Example: Citizen sensor agents, bystander agents, AED agents, volunteer responder agents and ambulance agents collaborate in SPADE to detect collapses, corroborate evidence and allocate responders and equipment. The scenario could model sensor false alarms, responder availability and bystander behaviour so students can design verification thresholds, decentralised dispatch and step-by-step guidance protocols that are safe and privacy-aware.

General instruction for all topics:

Each topic should be implemented using SPADE and is deliberately presented as open-ended: students must choose system norms, agent granularity, message schemas, negotiation and market mechanisms, verification and trust models, interaction protocols, and failure-handling strategies. For every submission include agent descriptions, message formats, failure modes, performance metrics, and a justification of design choices. Evaluate the system under at least three contrasting test scenarios and report results, trade-offs and limitations.

Deliverables:

- Source Code: SPADE-based multi-agent system with clear instructions to run simulations.
- Documentation: Design rationale, agent descriptions, message schemas, and protocol specifications, to be included in the final presentation; code documentation.
- Performance Report: Results from at least three test scenarios, metric analysis, and discussion of trade-offs.
- Visualisation/Demo: Interactive or recorded visualisation showing agent interactions and key metrics; presentation and accompanying slide-deck.
- Optional: Unit tests or automated scenario scripts to reproduce experiments.